$\Rightarrow \quad cx > acy \Rightarrow cx > abz \quad (\because cy = bz)$

$\Rightarrow \quad cx \neq abz \ i.e. \ cx \ \delta \ abz.$

44. (*a*) : $a^2x \ \alpha \ byz, czx \ \alpha \ b^2y$ and $c^2z \ \alpha \ axy \Rightarrow a^2x = byz, czx = b^2y, \ c^2z = axy.$

$czx = b^2y \Rightarrow c^2zx = cb^2y \Rightarrow axyx = cb^2y \quad (\because c^2z = axy)$

$\Rightarrow ax^2 = cb^2.$

Now, $a^2x = byz \Rightarrow a^2x^2 = bxyz \Rightarrow a.ax^2 = bxyz$

$\Rightarrow acb^2 = bxyz \quad (\because ax^2 = cb^2)$

$\Rightarrow abc = xyz \ i.e. \ abc \ \alpha \ xyz.$

45. (*a*) : $A + B > C + D, \ C + D > B + E, \ B + E = 2C$

$\Rightarrow A + B > B + E, \ B + E = 2C \Rightarrow A + B > 2C.$

46. (*d*) : $A + D > C + E \Rightarrow A + D > (2B - D) + E \quad (\because C + D = 2B)$

$\Rightarrow A + D > (B + E) + (B - D)$

$\Rightarrow A + D > (C + D) + (B - D)$

$\Rightarrow A + D > B + C.$

47. (*b*) : Clearly, time taken $= \dfrac{\text{sum of lengths of two trains}}{\text{total speed of two trains}}$

$= \dfrac{L_1 + L_2}{V_1 + V_2} = (L_1 \text{"} L_2) \ @ \ (V_1 \text{"} V_2).$

48. (*b*) : Total fare = B + 15% of B + 2% of B + 200

$= B + \dfrac{B \times 15}{100} + \dfrac{B \times 2}{100} + 200$

$= B \text{"} (B \bullet 15) \ @ \ 100 \text{"} (B \bullet 2) \ @ \ 100 \text{"} 200.$

49. (*c*) : Profit percentage $= \dfrac{S - (C + L + T)}{C + L + T} \times 100$

$= \{S' \ (C \text{"} L \text{"} \ T)\} \ @ \ (C \text{"} L' \ T) \bullet 100$

50. (*e*) : Clearly, total marks $= (T - 2) \times 2 + \dfrac{4K}{3} + 5 \times 2$

$= (T' 2) \bullet 2 \text{"} 4 \bullet K \ @ \ 3 \text{"} 5 \bullet 2.$

51. (*b*) : Marks out of 150 in first periodical = P.

Marks out of 100 in first periodical $= \left(\dfrac{P}{150} \times 100\right).$

Marks out of 180 in second periodical = T.

Marks out of 100 in second periodical $= \left(\dfrac{T}{180} \times 100\right).$

Marks out of 400 in final examination = M.

Marks out of 100 in final examination $= \left(\dfrac{M}{400} \times 100\right).$

$\therefore$ Total marks

$= \left[10\% \text{ of } \left(\dfrac{P}{150} \times 100\right)\right] + \left[15\% \text{ of } \left(\dfrac{T}{180} \times 100\right)\right] + \left[75\% \text{ of } \left(\dfrac{M}{400} \times 100\right)\right]$

$= \left[\dfrac{10}{100} \text{ of } \left(\dfrac{P}{150} \times 100\right)\right] + \left[\dfrac{15}{100} \text{ of } \left(\dfrac{T}{180} \times 100\right)\right] + \left[\dfrac{75}{100} \text{ of } \left(\dfrac{M}{400} \times 100\right)\right]$

$= \left(\dfrac{P}{150} \times 10\right) + \left(\dfrac{T}{180} \times 15\right) + \left(\dfrac{M}{400} \times 75\right)$

$= (P \ @ \ 150 \bullet 10) \text{"} (T \ @ \ 180 \bullet 15) \text{"} (M \ @ \ 400 \bullet 75).$

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13. LOGICAL SEQUENCE OF WORDS

In this type of questions, a group of words is given. The candidate is required to arrange these words in a meaningful order such as the sequence of occurrence of events, sequence from a part to the whole, sequence of increasing/decreasing size, value, intensity etc., and then choose the correct sequence accordingly.

Ex. 1. ***Arrange the following in a meaningful sequence :***

1. Consultation 2. Illness 3. Doctor
4. Treatment 5. Recovery

(*a*) 2,3,1,4,5 (*b*) 2,3,4,1,5 (*c*) 4,3,1,2,5 (*d*) 5,1,4,3,2

Sol. We know that **illness** occurs first. One then goes to the **doctor** and after **consultation** with him, undergoes **treatment** to finally attain **recovery.**

Thus, the correct order is 2, 3, 1, 4, 5.

Hence, the answer is (*a*).

Ex. 2. ***Arrange the following in a logical order :***

1. Euphoria 2. Happiness 3. Ambivalence
4. Ecstasy 5. Pleasure

(*a*) 4,1,3,2,5 (*b*) 3,2,5,1,4 (*c*) 2,1,3,4,5 (*d*) 1,4,2,5,3

Sol. All the given words stand for 'Joy', but the intensity increases in the order — Ambivalence, Happiness, Pleasure, Euphoria, Ecstasy.

Thus, the correct order is 3, 2, 5, 1, 4.

Hence, the answer is (*b*).

Ex. 3. ***Arrange the following in a meaningful order, from particular to general :***

1. Family 2. Community 3. Member
4. Locality 5. Country

(*a*) 3,1,2,4,5 (*b*) 3,1,2,5,4 (*c*) 3,1,4,2,5 (*d*) 3,1,4,5,2

(Asstt. Grade, 1996)

Sol. Clearly, a **member** is a part of a **family,** which in turn is a part of **community.** The community lives in a **locality** which lies within a **country.**

Thus, the correct order is 3, 1, 2, 4, 5.

Hence, the answer is (*a*).

EXERCISE 13

Directions (*Questions 1 to 33*) : ***In each of the following questions, arrange the given words in a meaningful sequence and then choose the most appropriate sequence from amongst the alternatives provided below each question.***

1. 1. Birth 2. Death 3. Funeral
4. Marriage 5. Education

(Asstt. Grade, 1995)

(*a*) 4,5,3,1,2 (*b*) 2,3,4,5,1 (*c*) 1,5,4,2,3 (*d*) 1,3,4,5,2

2. 1. Site 2. Plan 3. Rent
4. Money 5. Building

(Central Excise, 1996)

(*a*) 4,1,2,5,3 (*b*) 3,4,2,5,1 (*c*) 2,3,5,1,4 (*d*) 1,2,3,5,4

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3. 1. Table 2. Tree 3. Wood
4. Seed 5. Plant **(C.B.I. 1994)**
(*a*) 4,5,3,2,1 (*b*) 4,5,2,3,1 (*c*) 1,3,2,4,5 (*d*) 1,2,3,4,5

4. 1. College 2. Child 3. Salary
4. School 5. Employment **(Central Excise, 1992)**
(*a*) 1,2,4,3,5 (*b*) 2,4,1,5,3 (*c*) 4,1,3,5,2 (*d*) 5,3,2,1,4

5. 1. Reading 2. Composing 3. Writing 4. Printing
(*a*) 1,3,2,4 (*b*) 2,3,4,1 (*c*) 3,1,2,4 (*d*) 3,2,4,1
(Railways, 1995)

6. 1. Cutting 2. Dish 3. Vegetable
4. Market 5. Cooking **(Asstt. Grade, 1994)**
(*a*) 1,2,4,5,3 (*b*) 3,2,5,1,4 (*c*) 4,3,1,5,2 (*d*) 5,3,2,1,4

7. 1. Income 2. Status 3. Education
4. Well-being 5. Job **(S.S.C. 1993)**
(*a*) 1,3,2,5,4 (*b*) 1,2,5,3,4 (*c*) 3,1,5,2,4 (*d*) 3,5,1,2,4

8. 1. Milky way 2. Sun 3. Moon
4. Earth 5. Stars **(C.B.I. 1993)**
(*a*) 4,3,2,5,1 (*b*) 3,4,2,5,1 (*c*) 2,3,4,5,1 (*d*) 1,4,3,2,5

9. 1. Sea 2. Rivulet 3. Ocean
4. River 5. Glacier **(I. Tax & Central Excise, 1995)**
(*a*) 5,4,3,2, 1 (*b*) 5,4,2,3,1 (*c*) 5,2,4,1,3 (*d*) 5,2,1,3,4

10. 1. Poverty 2. Population 3. Death
4. Unemployment 5. Disease **(S.S.C. 1996)**
(*a*) 3,4,2,5,1 (*b*) 2,4,1,5,3 (*c*) 2,3,4,5,1 (*d*) 1,2,3,4,5

11. 1. Yarn 2. Plant 3. Saree
4. Cotton 5. Cloth **(Asstt. Grade, 1996)**
(*a*) 2,4,1,5,3 (*b*) 2,4,3,5,1 (*c*) 2,4,5,1,3 (*d*) 2,4,5,3,1

12. 1. Puberty 2. Adulthood 3. Childhood
4. Infancy 5. Senescence **(U.D.C. 1995)**
(*a*) 5,2,3,4,1 (*b*) 4,3,2,1,5 (*c*) 4,3,1,2,5 (*d*) 2,4,3,1,5

13. 1. Windows 2. Walls 3. Floor
4. Foundation 5. Roof 6. Room **(C.B.I. 1995)**
(*a*) 4,5,3,2,1,6 (*b*) 4,3,5,6,2,1 (*c*) 4,2,1,5,3,6 (*d*) 4,1,5,6,2,3

14. 1. Post-box 2. Letter 3. Envelope
4. Delivery 5. Clearance
(*a*) 3,2,4,5,1 (*b*) 3,2,1,5,4 (*c*) 3,2,1,4,5 (*d*) 2,3,1,4,5

15. 1. Key 2. Door 3. Lock
4. Room 5. Switch on **(Asstt. Grade, 1995)**
(*a*) 5,1,2,4,3 (*b*) 4,2,1,5,3 (*c*) 1,2,3,5,4 (*d*) 1,3,2,4,5

16. 1. Gold 2. Iron 3. Sand
4. Platinum 5. Diamond **(Asstt. Grade, 1997)**
(*a*) 2,4,3,5,1 (*b*) 3,2,1,5,4 (*c*) 4,5,1,3,2 (*d*) 5,4,3,2,1

17. 1. Cut 2. Put on 3. Mark
4. Measure 5. Tailor **(Central Excise, 1994)**
(*a*) 4,3,1,5,2 (*b*) 3,1,5,4,2 (*c*) 2,4,3,1,5 (*d*) 1,3,2,4,5

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18. 1. Rainbow 2. Rain 3. Sun
4. Happy 5. Child **(S.S.C. 1993)**
(*a*) 2,1,4,3,5 (*b*) 2,3,1,5,4 (*c*) 4,2,3,5,1 (*d*) 4,5,1,2,3

19. 1. Study 2. Job 3. Examination
4. Earn 5. Apply **(I. Tax & Central Excise, 1992)**
(*a*) 1,2,3,4,5 (*b*) 1,3,2,5,4 (*c*) 1,3,5,4,2 (*d*) 1,3,5,2,4

20. 1. Shoulder 2. Wrist 3. Elbow
4. Palm 5. Finger
(*a*) 5,4,2,3,1 (*b*) 3,4,5,2,1 (*c*) 3,1,4,2,5 (*d*) 2,4,5,3,1

21. 1. Frog 2. Eagle 3. Grasshopper
4. Snake 5. Grass **(S.S.C. 1996)**
(*a*) 5,3,4,2,1 (*b*) 5,3,1,4,2 (*c*) 3,4,2,5,1 (*d*) 1,3,5,2,4

22. 1. Punishment 2. Prison 3. Arrest
4. Crime 5. Judgement **(Asstt. Grade, 1996)**
(*a*) 5,1,2,3,4 (*b*) 4,3,5,2,1 (*c*) 4,3,5,1,2 (*d*) 2,3,1,4,5

23. 1. Child 2. Job 3. Marriage
4. Infant 5. Education **(S.S.C. 1993)**
(*a*) 1,3,5,2,4 (*b*) 3,5,2,1,4 (*c*) 4,1,3,5,2 (*d*) 4,1,5,2,3

24. 1. Mother 2. Child 3. Milk
4. Cry 5. Smile **(Asstt. Grade, 1995)**
(*a*) 1,5,2,4,3 (*b*) 2,4,1,3,5 (*c*) 2,4,3,1,5 (*d*) 3,2,1,5,4

25. 1. Travel 2. Destination 3. Payment
4. Berth/Seat number 5. Reservation
6. Availability of berth/seat for reservation
(*a*) 6,2,5,4,3,1 (*b*) 5,3,4,1,6,2 (*c*) 2,6,3,5,4,1 (*d*) 1,2,5,4,3,6
(Central Excise, 1994)

26. 1. Curd 2. Grass 3. Butter
4. Milk 5. Cow **(Asstt. Grade, 1994)**
(*a*) 5,2,4,1,3 (*b*) 5,2,3,4,1 (*c*) 4,2,5,3,1 (*d*) 2,5,4,3,1

27. 1. Elephant 2. Cat 3. Mosquito
4. Tiger 5. Whale **(Asstt. Grade, 1996)**
(*a*) 1,3,5,4,2 (*b*) 2,5,1,4,3 (*c*) 3,2,4,1,5 (*d*) 5,3,1,2,4

28. 1. Probation 2. Interview 3. Selection
4. Appointment 5. Advertisement 6. Application
(*a*) 5,6,2,3,4,1 (*b*) 5,6,3,2,4,1 (*c*) 5,6,4,2,3,1 (*d*) 6,5,4,2,3,1

29. 1. District 2. Village 3. State 4. Town
(*a*) 2,1,4,3 (*b*) 2,3,4,1 (*c*) 2,4,1,3 (*d*) 3,2,1,4
(S.S.C. 1995)

30. 1. Index 2. Contents 3. Title
4. Chapters 5. Introduction
(*a*) 3,2,5,1,4 (*b*) 2,3,4,5,1 (*c*) 5,1,4,2,3 (*d*) 3,2,5,4,1

31. 1. Country 2. Furniture 3. Forest
4. Wood 5. Trees **(S.S.C. 1993)**
(*a*) 1,3,5,4,2 (*b*) 1,4,3,2,5 (*c*) 2,4,3,1,5 (*d*) 5,2,3,1,4

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32. 1. Protect 2. Pressure 3. Relief
4. Rain 5. Flood **(I. Tax & Central Excise, 1994)**
(*a*) 2,4,3,1,5 (*b*) 2,4,5,1,3 (*c*) 2,5,4,1,3 (*d*) 3,2,4,5,1

33. 1. Andhra Pradesh 2. Universe 3. Tirupathi
4. World 5. India **(Asstt. Grade, 1995)**
(*a*) 1,5,3,2,4 (*b*) 2,1,3,5,4 (*c*) 3,1,5,4,2 (*d*) 5,4,2,1,3

34. Arrange the following in a meaningful order : **(C.B.I. 1997)**
Doctor, Fever, Medicine, Medical shop
(*a*) Medicine, Doctor, Medical shop, Fever
(*b*) Doctor, Medical shop, Medicine, Fever
(*c*) Fever, Doctor, Medical shop, Medicine
(*d*) Medical shop, Medicine, Fever, Doctor

35. Which would be the proper order of the following (in ascending order) ?
1. Trillion 2. Thousand 3. Billion
4. Hundred 5. Million **(S.S.C. 1993)**
(*a*) 1,2,4,3,5 (*b*) 1,5,3,2,4 (*c*) 4,2,3,5,1 (*d*) 4,2,5,3,1

36. Which of the following number sequences represents a correct sequence from a part to the whole ?
1. Caste 2. Family 3. Newly married couple
4. Clan 5. Species **(C.B.I. 1993)**
(*a*) 2,3,1,4,5 (*b*) 3,2,1,4,5 (*c*) 3,4,5,1,2 (*d*) 4,5,3,2,1

37. Arrange the following items from general to particular :
1. Animal 2. Feline 3. Leopard
4. Mammal 5. Vertebrate 6. Cat
(*a*) 1,5,4,2,3,6 (*b*) 1,4,3,2,5,6 (*c*) 1,3,5,4,2,6 (*d*) 1,2,3,4,5,6

38. Which number sequence of the following represents a correct sequence from small to big ?
1. Bungalow 2. Flat 3. Cottage
4. House 5. Palace 6. Mansion
(*a*) 3,2,1,4,6,5 (*b*) 3,2,4,1,5,6 (*c*) 3,2,4,1,6,5 (*d*) 5,6,4,1,2,3

ANSWERS

1. (*c*)	**2.** (*a*)	**3.** (*b*)	**4.** (*b*)	**5.** (*d*)	**6.** (*c*)	**7.** (*d*)	**8.** (*b*)	**9.** (*c*)	**10.** (*b*)
11. (*a*)	**12.** (*c*)	**13.** (*c*)	**14.** (*b*)	**15.** (*d*)	**16.** (*b*)	**17.** (*a*)	**18.** (*b*)	**19.** (*d*)	**20.** (*a*)
21. (*b*)	**22.** (*c*)	**23.** (*d*)	**24.** (*b*)	**25.** (*c*)	**26.** (*a*)	**27.** (*c*)	**28.** (*a*)	**29.** (*c*)	**30.** (*d*)
31. (*a*)	**32.** (*b*)	**33.** (*c*)	**34.** (*c*)	**35.** (*d*)	**36.** (*b*)	**37.** (*a*)	**38.** (*c*)		

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14. ARITHMETICAL REASONING

TYPE 1 : CALCULATION-BASED PROBLEMS

Ex. 1. In a chess tournament each of six players will play every other player exactly once. How many matches will be played during the tournament ?

(*a*) 12 (*b*) 15 (*c*) 30 (*d*) 36

(C.B.I. 1995)

Sol. Clearly, we will consider the following matches :

(*i*) matches of first player with other 5 players;

(*ii*) matches of second player with 4 players other than the first player;

(*iii*) matches of third player with 3 players other than the first and second players;

(*iv*) matches of fourth player with 2 players other than the first three players; and

(*v*) match of fifth player with the sixth one.

So, number of matches played during the tournament = 5 + 4 + 3 + 2 + 1 = 15.

Ex. 2. A man has a certain number of small boxes to pack into parcels. If he packs 3, 4, 5 or 6 in a parcel, he is left with one over; if he packs 7 in a parcel, none is left over. What is the number of boxes, he may have to pack ?

(*a*) 106 (*b*) 301 (*c*) 309 (*d*) 400

Sol. Clearly, the required number would be such that it leaves a remainder of 1 when divided by 3, 4, 5 or 6 and no remainder when divided by 7. Such a number is 301.

Hence, the answer is (*b*).

Ex. 3. A, B, C and D play a game of cards. A says to B, "If I give you 8 cards, you will have as many as C has and I shall have 3 less than what C has. Also, if I take 6 cards from C, I shall have twice as many as D has." If B and D together have 50 cards, how many cards has A got ? **(Hotel Management, 1997)**

(*a*) 40 (*b*) 37 (*c*) 27 (*d*) 23

Sol. Clearly, we have :

$B + 8 = C$ ...(*i*) $A - 8 = C - 3$ ...(*ii*)

$A + 6 = 2D$ ...(*iii*) $B + D = 50$ ...(*iv*)

Putting $C = A - 5$ from (*ii*) into (*i*), we have :

$B + 8 = A - 5$ or $A - B = 13$ ...(*v*)

Putting $D = 50 - B$ from (*iv*) into (*iii*), we have :

$A + 6 = 100 - 2B$ or $A + 2B = 94$ ...(*vi*)

Solving (*v*) and (*vi*), we get $B = 27$ and $A = 40$.

$\therefore$ A has 40 cards.

Hence, the answer is (*a*).

Ex. 4. In a group of cows and hens, the number of legs are 14 more than twice the number of heads. The number of cows is

(*a*) 5 (*b*) 7 (*c*) 10 (*d*) 12

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Sol. Let the number of cows be x and the number of hens be y. Then,
number of legs in the group $= 4x + 2y$.
number of heads in the group $= x + y$.
So, $4x + 2y = 2(x + y) + 14$ or $4x + 2y = 2x + 2y + 14$ or $2x = 14$ or $x = 7$.
$\therefore$ Number of cows = 7.
Hence, the answer is (*b*).

Ex. 5. A worker may claim Rs 15 for each km which he travels by taxi and Rs 5 for each km which he drives his own car. If in one week he claimed Rs 500 for travelling 80 km how many kms did he travel by taxi ?
(*a*) 10 (*b*) 20 (*c*) 30 (*d*) 40

Sol. Let the distance covered by taxi be x km.
Then, distance covered by car $= (80 - x)$ km.
$15x + 5(80 - x) = 500$ or $15x + 400 - 5x = 500$ or $10x = 100$ or $x = 10$.
$\therefore$ Distance covered by taxi = 10 km.
Hence, the answer is (*a*).

TYPE 2 : DATA-BASED QUESTIONS

Ex. 6. The following questions are based on the given data for an examination.

(A)	Candidates appeared	10500
(B)	Passed in all the five subjects	5685
(C)	Passed in three subjects only	1498
(D)	Passed in two subjects only	1250
(E)	Passed in one subject only	835
(F)	Failed in English only	78
(G)	Failed in Maths only	275
(H)	Failed in Physics only	149
(I)	Failed in Chemistry only	147
(J)	Failed in Biology only	221

Q. 1. How many candidates failed in all the subjects ?
(*a*) 4815 (*b*) 3317 (*c*) 2867 (*d*) 362

Sol. Clearly, candidates failed in all the subjects
= (Candidates appeared) – (Candidates failed in 1, 2, 3 or 5 subjects + Candidates failed in 1 subject only)
= 10500 – (5685 + 1498 + 1250 + 835 + 78 + 275 + 149 + 147 + 221)
= 10500 – 10138 = 362.
Hence, the answer is (*d*).

Q. 2. How many candidates passed at least in four subjects ?
(*a*) 6555 (*b*) 5685 (*c*) 1705 (*d*) 870

Sol. Candidates passed at least in four subjects
= (Candidates passed in 4 subjects) + (Candidates passed in all 5 subjects)
= (Candidates failed in only 1 subject) + (Candidates passed in all 5 subjects)
= (78 + 275 + 149 + 147 + 221) + 5685
= 870 + 5685 = 6555.
Hence, the answer is (*a*).

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Q. 3. How many candidates failed because of having failed in four or less subjects ?

(*a*) 4815 (*b*) 4453 (*c*) 3618 (*d*) 2368

Sol. Candidates failed in four or less subjects

= (Candidates failed in only 1 subject) + (Candidates failed in only 2 subjects) + (Candidates failed in only 3 subjects) + (Candidates failed in only 4 subjects)

= (Candidates failed in only 1 subject) + (Candidates passed in only 3 subjects) + (Candidates passed in only 2 subjects) + (Candidates passed in only 1 subject)

$= (78 + 275 + 149 + 147 + 221) + 1498 + 1250 + 835$

$= 4453.$

Hence, the answer is (*b*).

TYPE 3 : PROBLEMS ON AGES

Ex. 7. Reena is twice as old as Sunita. Three years ago, she was three times as old as Sunita. How old is Reena now ? **(I. Tax & Central Excise, 1995)**

(*a*) 6 years (*b*) 7 years (*c*) 8 years (*d*) 12 years

Sol. Let Sunita's present age be x years.

Then, Reena's present age = $2x$ years.

Three years ago, Sunita's age = $(x - 3)$ and Reena's age = $(2x - 3)$.

So, $(2x - 3) = 3(x - 3)$ or $2x - 3 = 3x - 9$ or $x = 6$.

$\therefore$ Reena's present age = $2x = 12$ years.

Hence, the answer is (*d*).

Ex. 8. The age of a father is twice that of the elder son. Ten years hence the age of the father will be three times that of the younger son. If the difference of ages of the two sons is 15 years, the age of the father is

(*a*) 50 years (*b*) 55 years (*c*) 60 years (*d*) 70 years

Sol. Let the age of the elder son be x.

Then, age of younger son = $(x - 15)$; and

age of the father = $2x$.

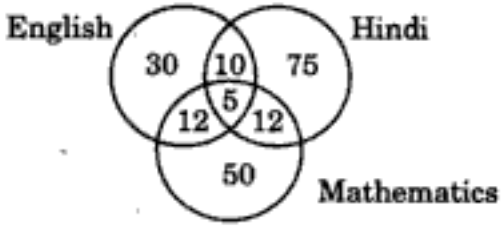
So, $2x + 10 = 3(x - 15 + 10)$ or $2x + 10 = 3x - 15$ or $x = 25$.

$\therefore$ Father's age = $2x = 50$ years.

Hence, the answer is (*a*).

TYPE 4 : VENN-DIAGRAM BASED QUESTIONS

Ex. 9. Consider the diagram given below : **(I.A.S. 1994)**



Five hundred candidates appeared in an examination comprising of tests in English, Hindi and Mathematics. The diagram gives the number of candidates who failed in different tests. What is the percentage of candidates who failed in at least two subjects ?

(*a*) 0.078 (*b*) 1.0 (*c*) 6.8 (*d*) 7.8

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Sol. Clearly, number of candidates who failed in at least two subjects
= number of candidates who failed in two or more subjects
= (10 + 12 + 12 + 5) = 39.

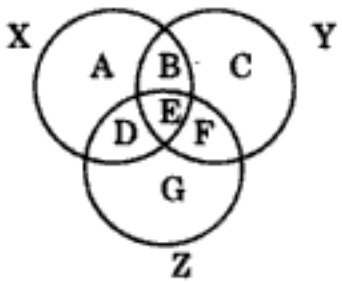
$\therefore$ Required percentage $= \left(\frac{39}{500} \times 100\right)\% = 7.8\%$.

Hence, the answer is (*d*).

Ex. 10. In a group of persons travelling in a bus, 6 persons can speak Tamil, 15 can speak Hindi and 6 can speak Gujarati. In that group, none can speak any other language. If 2 persons in the group can speak two languages and one person can speak all the three languages, then how many persons are there in the group ? **(I.A.S. 1997)**

(*a*) 21 (*b*) 22 (*c*) 23 (*d*) 24

Sol. Let circles X, Y, and Z represent persons who can speak Tamil, Hindi and Gujarati respectively.



Tamil-speaking persons = A + B + D + E = 6 ...(*i*)
Hindi-speaking persons = B + C + E + F = 15 ...(*ii*)
Gujarati-speaking persons = D + E + F + G = 6 ...(*iii*)
Persons speaking 2 languages = B + D + F = 2 ...(*iv*)
Persons speaking all 3 languages = E = 1 ...(*v*)
Clearly, we have : A + B + D = 5 ...(*vi*)
B + C + F = 14 ...(*vii*)
D + F + G = 5 ...(*viii*)
B + D + F = 2 ...(*ix*)
Subtracting (*ix*) from (*vi*), we get :
A − F = 3 ...(*x*)
Adding (*vii*) and (*viii*), we get :
B + C + D + 2F + G = 19 ...(*xi*)
Adding (*x*) and (*xi*), we get :
A + B + C + D + F + G = 22
or A + B + C + D + E + F + G = 23. ($\because$ E = 1)
$\therefore$ Total number of persons = 23.
Hence, the answer is (*c*).

EXERCISE 14

1. A shepherd had 17 sheep. All but nine died. How many was he left with ?

(*a*) Nil (*b*) 8 (*c*) 9 (*d*) 17

(Railways, 1995)

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2. A bird shooter was asked how many birds he had in the bag. He replied that there were all sparrows but six, all pigeons but six, and all docks but six. How many birds had he in all ?

(*a*) 9 (*b*) 18 (*c*) 27 (*d*) 36

3. What is the smallest number of ducks that could swim in this formation — two ducks in front of a duck, two ducks behind a duck and a duck between two ducks ?

(*a*) 3 (*b*) 5 (*c*) 7 (*d*) 9

4. A group of 1200 persons consisting of captains and soldiers is travelling in a train. For every 15 soldiers there is one captain. The number of captains in the group is **(Hotel Management, 1992)**

(*a*) 85 (*b*) 80 (*c*) 75 (*d*) 70

5. Aruna cut a cake into two halves and cuts one half into smaller pieces of equal size. Each of the small pieces is twenty grams in weight. If she has seven pieces of the cake in all with her, how heavy was the original cake ? **(L.I.C. 1994)**

(*a*) 120 grams (*b*) 140 grams (*c*) 240 grams
(*d*) 280 grams (*e*) None of these

6. First bunch of bananas has $\frac{1}{4}$ again as many bananas as a second bunch. If the second bunch has 3 bananas less than the first bunch, then the number of bananas in the first bunch are **(S.C.R.A. 1996)**

(*a*) 9 (*b*) 10 (*c*) 12 (*d*) 15

7. At the end of a business conference the ten people present all shake hands with each other once. How many handshakes will there be altogether ? **(M.B.A. 1997)**

(*a*) 20 (*b*) 45 (*c*) 55 (*d*) 90

8. A student got twice as many sums wrong as he got right. If he attempted 48 sums in all, how many did he solve correctly ? **(M.B.A. 1994)**

(*a*) 12 (*b*) 16 (*c*) 24 (*d*) 18

9. The number of boys in a class is three times the number of girls. Which one of the following numbers cannot represent the total number of children in the class ? **(S.C.R.A. 1993)**

(*a*) 48 (*b*) 44 (*c*) 42 (*d*) 40

10. A placed three sheets with two carbons to get two extra copies of the original. Then he decided to get more carbon copies and folded the paper in such a way that the upper half of the sheets were on top of the lower half. Then he typed. How many carbon copies did he get ?

(*a*) 1 (*b*) 2 (*c*) 3 (*d*) 4

11. A motorist knows four different routes from Bristol to Birmingham. From Birmingham to Sheffield he knows three different routes and from Sheffield to Carlisle he knows two different routes. How many routes does he know from Bristol to Carlisle ?

(*a*) 4 (*b*) 8 (*c*) 12 (*d*) 24

12. In a class, there are 18 boys who are over 160 cm tall. If these constitute three-fourths of the boys and the total number of boys is two-thirds of the total number of students in the class, what is the number of girls in the class ? **(I.A.S. 1992)**

(*a*) 6 (*b*) 12 (*c*) 18 (*d*) 24

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13. A bus starts from city X. The number of women in the bus is half of the number of men. In city Y, 10 men leave the bus and five women enter. Now, number of men and women is equal. In the beginning, how many passengers entered the bus ? **(I. Tax & Central Excise, 1995)**

(*a*) 15 (*b*) 30 (*c*) 36 (*d*) 45

14. In a class, 20% of the members own only two cars each, 40% of the remaining own three cars each and the remaining members own only one car each. Which of the following statements is definitely true from the given statements ?

(*a*) Only 20% of the total members own three cars each.

(*b*) 48% of the total members own only one car each.

(*c*) 60% of the total members own at least two cars each.

(*d*) 80% of the total members own at least one car.

(*e*) None of these **(Bank P.O. 1998)**

15. Between two book-ends in your study are displayed your five favourite puzzle books. If you decide to arrange the five books in every possible combination and moved just one book every minute, how long would it take you ?

(*a*) 1 hour (*b*) 2 hours (*c*) 3 hours (*d*) 4 hours

16. I have a few sweets to be distributed. If I keep 2, 3 or 4 in a pack, I am left with one sweet. If I keep 5 in a pack, I am left with none. What is the minimum number of sweets I can have to pack and distribute ? **(Assistant Grade, 1992)**

(*a*) 25 (*b*) 37 (*c*) 54 (*d*) 65

17. Mr. Johnson was to earn £ 300 and a free holiday for seven weeks' work. He worked for only 4 weeks and earned £ 30 and a free holiday. What was the value of the holiday ?

(*a*) £ 300 (*b*) £ 330 (*c*) £ 360 (*d*) £ 420

18. In a cricket match, five batsmen A, B, C, D and E scored an average of 36 runs. D scored 5 more than E; E scored 8 fewer than A; B scored as many as D and E combined; and B and C scored 107 between them. How many runs did E score ?

(*a*) 62 (*b*) 45 (*c*) 28 (*d*) 20

19. Mac has £ 3 more than Ken, but then Ken wins on the horses and trebles his money, so that he now has £ 2 more than the original amount of money that the two boys had between them. How much money did Mac and Ken have between them before Ken's win ?

(*a*) £ 9 (*b*) £ 11 (*c*) £ 13 (*d*) £ 15

20. Robin says, "If Jai gives me Rs 40, he will have half as much as Atul, but if Atul gives me Rs 40, then the three of us will all have the same amount." What is the total amount of money that Robin, Jai and Atul have between them ?

(*a*) Rs 240 (*b*) Rs 320 (*c*) Rs 360 (*d*) Rs 420

21. A, B, C, D and E play a game of cards. A says to B, "If you give me three cards, you will have as many as E has and if I give you three cards, you will have as many as D has." A and B together have 10 cards more than what D and E together have. If B has two cards more than what C has and the total number of cards be 133, how many cards does B have ? **(Hotel Management, 1995)**

(*a*) 22 (*b*) 23 (*c*) 25 (*d*) 35

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22. A, B, C, D and E play a game of cards. A says to B, "If you give me 3 cards, you will have as many as I have at this moment while if D takes 5 cards from you, he will have as many as E has." A and C together have twice as many cards as E has. B and D together also have the same number of cards as A and C taken together. If together they have 150 cards, how many cards has C got ?

(*a*) 28 (*b*) 29 (*c*) 31 (*d*) 35

(C.A.T. 1997)

23. In a caravan in addition to 50 hens, there are 45 goats and 8 camels with some keepers. If the total number of feet be 224 more than the number of heads in the caravan, the number of keepers is **(Hotel Management, 1995)**

(*a*) 5 (*b*) 8 (*c*) 10 (*d*) 15

24. A certain number of horses and an equal number of men are going somewhere. Half of the owners are on their horses' back while the remaining ones are walking along leading their horses. If the number of legs walking on the ground is 70, how many horses are there ? **(Railways, 1998)**

(*a*) 10 (*b*) 12 (*c*) 14 (*d*) 16

25. At a farm, there are hens, cows and bullocks, and keepers to look after them. There are 69 heads less than legs; the number of cows is double of that of the bullocks; the number of cows and hens is the same and there is one keeper per ten birds and cattle. The total number of hens plus cows and bullocks and their keepers does not exceed 50. How many cows are there ?

(*a*) 10 (*b*) 12 (*c*) 14 (*d*) 16

26. In a certain office, $\frac{1}{3}$ of the workers are women, $\frac{1}{2}$ of the women are married and $\frac{1}{3}$ of the married women have children. If $\frac{3}{4}$ of the men are married and $\frac{2}{3}$ of the married men have children, what part of workers are without children ?

(*a*) $\frac{5}{18}$ (*b*) $\frac{4}{9}$ (*c*) $\frac{11}{18}$ (*d*) $\frac{17}{36}$

27. In an examination, a student scores 4 marks for every correct answer and loses 1 mark for every wrong answer. If he attempts all 75 questions and secures 125 marks, the number of questions he attempts correctly, is

(*a*) 35 (*b*) 40 (*c*) 42 (*d*) 46

Directions (*Questions 28 to 32*) : *The following questions are based on the information given below :* **(Hotel Management, 1997)**

Data on 450 candidates, who took an examination in Social Sciences, Mathematics and Science is given below :

Passed in all the subjects	167
Failed in all the subjects	60
Failed in Social Sciences	175
Failed in Mathematics	199
Failed in Science	191
Passed in Social Sciences only	62
Passed in Mathematics only	48
Passed in Science only	52

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28. How many failed in Social Sciences only ?
(*a*) 15 (*b*) 21 (*c*) 30 (*d*) 42

29. How many failed in one subject only ?
(*a*) 152 (*b*) 144 (*c*) 61 (*d*) 56

30. How many passed in Mathematics and at least one more subject ?
(*a*) 210 (*b*) 203 (*c*) 170 (*d*) 94

31. How many failed in two subjects only ?
(*a*) 56 (*b*) 61 (*c*) 152 (*d*) 162

32. How many passed at least in one subject ?
(*a*) 450 (*b*) 390 (*c*) 304 (*d*) 167

33. A is three times as old as B. C was twice as old as A four years ago. In four years' time, A will be 31. What is the present age of B and C ?
(*a*) 9, 46 (*b*) 9, 50 (*c*) 10, 46 (*d*) 10, 50
(Assistant Grade, 1997)

34. A father tells his son, "I was of your present age when you were born." If the father is 36 now, how old was the boy 5 years back ? **(Railways, 1994)**
(*a*) 13 (*b*) 15 (*c*) 17 (*d*) 20

35. In a family, a couple has a son and a daughter. The age of the father is three times that of his daughter and the age of the son is half of his mother. The wife is 9 years younger to her husband and the brother is seven years older than his sister. What is the age of the mother ? **(I.A.S. 1998)**
(*a*) 40 years (*b*) 45 years (*c*) 50 years (*d*) 60 years

36. A father is now three times as old as his son. Five years back, he was four times as old as his son. The age of the son is **(I.A.S. 1993)**
(*a*) 12 (*b*) 15 (*c*) 18 (*d*) 20

37. Ravi's brother is 3 years senior to him. His father was 28 years of age when his sister was born while his mother was 26 years of age when he was born. If his sister was 4 years of age when his brother was born, what was the age of Ravi's father and mother respectively when his brother was born ?
(*a*) 32 years, 23 years (*b*) 32 years, 29 years
(*c*) 35 years, 29 years (*d*) 35 years, 33 years
(Hotel Management, 1995)

38. When Rahul was born, his father was 32 years older than his brother and his mother was 25 years older than his sister. If Rahul's brother is 6 years older than him and his mother is 3 years younger than his father, how old was Rahul's sister when he was born ? **(Hotel Management, 1997)**
(*a*) 7 years (*b*) 10 years (*c*) 14 years (*d*) 19 years

39. In a town, 65% people watched the news on television, 40% read a newspaper and 25% read a newspaper and watched the news on television also. What percent of the people neither watched the news on television nor read a newspaper ?
(*a*) 5 (*b*) 10 (*c*) 15 (*d*) 20

40. In a group of 15 people, 7 read French, 8 read English while 3 of them read none of these two. How many of them read French and English both ?
(*a*) 0 (*b*) 3 (*c*) 4 (*d*) 5
(I. Tax & Central Excise, 1995)

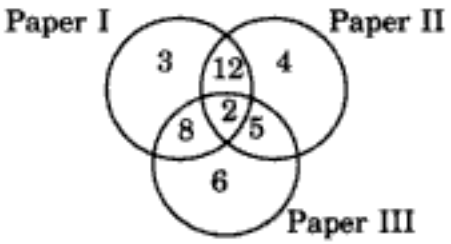
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41. There are 50 students admitted to a nursery class. Some students can speak only English and some can speak only Hindi. Ten students can speak both English and Hindi. If the number of students who can speak English is 21, then how many students can speak Hindi, how many can speak only Hindi and how many can speak only English ? **(I.A.S. 1998)**

(*a*) 39, 29 and 11 respectively (*b*) 37, 27 and 13 respectively

(*c*) 28, 18 and 22 respectively (*d*) 21, 11 and 29 respectively

42. Consider the Venn diagram given below : **(I.A.S. 1993)**



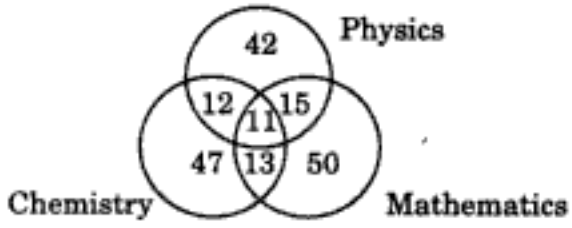
The number in the Venn diagram indicates the number of persons reading the newspapers. The diagram is drawn after surveying 50 persons. In a population of 10,000, how many can be expected to read at least two newspapers ?

(*a*) 5000 (*b*) 5400 (*c*) 6000 (*d*) 6250

43. Out of a total of 120 musicians in a club, 5% can play all the three instruments — guitar, violin and flute. It so happens that the number of musicians who can play any two and only two of the above instruments is 30. The number of musicians who can play the guitar alone is 40. What is the total number of those who can play violin alone or flute alone ? **(I.A.S. 1995)**

(*a*) 30 (*b*) 38 (*c*) 44 (*d*) 45

Directions (*Questions 44 to 46*) : *The diagram given below shows the number of students who got distinction in three subjects out of 500 students. Study the diagram carefully and answer the questions that follow.*



44. What is the percentage of students who got distinction in two subjects ?

(*a*) 8% (*b*) 9% (*c*) 10% (*d*) 12%

45. What is the percentage of students who got distinction ?

(*a*) 28% (*b*) 35% (*c*) 38% (*d*) 40%

46. The percentage of students with distinction marks in Mathematics is

(*a*) 17.8% (*b*) 18.6% (*c*) 19.2% (*d*) 20.6%

Directions (*Questions 47 to 49*) : *Study the information given below and answer the questions that follow :*

A publishing firm publishes newspapers A, B and C. In an effort to persuade advertisers to insert advertisements in these newspapers, the firm sends out the following statement to possible advertisers :

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A survey of representative sample of the whole population shows that —
Newspaper A is read by 26%;
Newspaper B is read by 25%;
Newspaper C is read by 14%;
Newspaper A and B are read by 11%;
Newspaper B and C are read by 10%;
Newspaper C and A are read by 9%;
Newspaper C only is read by 0%.

47. The percentage of readers who read all the three newspapers is
(*a*) 1 (*b*) 4 (*c*) 5 (*d*) 6

48. The percentage of readers who read A and B but not C, is
(*a*) 2 (*b*) 4 (*c*) 5 (*d*) 6

49. The percentage of readers who read at least one of the three newspapers is
(*a*) 40 (*b*) 50 (*c*) 60 (*d*) 65

50. A number of friends decided to go on a picnic and planned to spend Rs 96 on eatables. Four of them, however, did not turn up. As a consequence, the remaining ones had to contribute Rs 4 each extra. The number of those who attended the picnic was
(*a*) 8 (*b*) 12 (*c*) 16 (*d*) 24

ANSWERS

1. (*c*) : 'All but nine died' means 'All except nine died' *i.e.* nine sheep remained alive.

2. (*a*) : 'There were all sparrows but six' means that six birds were not sparrows but only pigeons and docks.
Similarly, number of sparrows + number of docks = 6
and number of sparrows + number of pigeons = 6.
This is possible when there are 3 sparrows, 3 pigeons and 3 docks *i.e.* 9 birds in all.

3. (*a*) : Clearly, the smallest such number is 3.
Three ducks can be arranged as shown along side to satisfy all the three given conditions.

D
D
D

4. (*c*) : Clearly, out of every 16 persons, there is one captain.

So, number of captains $= \frac{1200}{16} = 75$.

5. (*c*) : The seven pieces consist of 6 smaller equal pieces and one half cake piece.
Weight of each small piece = 20 g.
So, total weight of the cake = 2 × (20 × 6) = 240 g.

6. (*d*) : Let the number of bananas in the second bunch be x.

Then, number of bananas in the first bunch $= x + \frac{1}{4}x = \frac{5}{4}x$.

So, $\frac{5}{4}x - x = 3 \Rightarrow 5x - 4x = 12 \Rightarrow x = 12$.

$\therefore$ Number of bananas in first bunch $= \left(\frac{5}{4} \times 12\right) = 15$.

7. (*b*) : Clearly, total number of handshakes $= (9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1) = 45$.

8. (*b*) : Suppose the boy got x sums right and $2x$ sums wrong. Then,

$x + 2x = 48$ or $3x = 48$ or $x = 16$.

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